

Activity - 4

Hashing Problem

given input { 4371, 1323, 6173, 4199, 4344, 9679, 1989 } and a hash function $h(x) = x \pmod{10}$.

a) Separate Chaining hash Table

Index	Chain
0	-
1	4371
2	-
3	1323 → 6173
4	4344
5	-
6	-
7	-
8	-
9	4199 - 9679 → 1989

B) Open

Addressing - Linear

Bobing

giving

giving

$$h(i) = h(x) + i \text{ mod } 10$$

Where $i = 0, 1, 2, 3, \dots$

Insert one by one

4 3 7 1

$h = 1$

$\rightarrow$ index 1

1 3 2 3

$h = 3, i = 3$

6 1 7 3, $h = 3 \rightarrow$ occupied

then

$$(3+1) \text{ mod } 10 = 4$$

$i = 4$

4 1 9 9

$h = 9, i = 9$

4 3 4 4

$h = 4$

9 6 7 9

$h = 9 \rightarrow$ occupied

$i = 0$

1 9 8 9

~~$h = 9$~~ $h = 9 \rightarrow$ occupied

$i = 2$

Index	Value
0	9679
1	4371
2	1989
3	1323
4	6173
5	4344
6	-
7	-
8	-
9	4199

c) Open Addressing - Quadratic Probing

Insert

4371

$h=1 \rightarrow \text{index } 1$

1323

$h=3 \rightarrow \text{index } 3$

6173

$h=3 \rightarrow \text{occupied}$

$i=4$

4199

$h=9 \rightarrow \text{index } 9$

4344

$h=4 \rightarrow \text{occupied}$

index $\rightarrow 5$

9679

$h=9$

$\rightarrow \text{index } 0$

1989

$h = a$ occupied

index $\rightarrow 1$

For $i=2$:

$$(9 + 2^2) \text{ mod } 10 = (9 + 4) \text{ mod } 10 = 3$$

for $i=3$:

$$(9 + 3^2) = 18 \text{ mod } 10 = 8$$

$\rightarrow$ index 8

Index	Value
0	9679
1	4371
2	-
3	1323
4	6173
5	4344
6	-
7	-
8	1989
9	4199

d) Open Addressing - Double Hashing
Given

$$h_1(x) = x \text{ mod } 10$$

$$h_2(x) = 7 - (x \text{ mod } 7)$$

Formula:

$$h_i(x) = (h_1(x) + i \times h_2(x)) \text{ mod } 10$$

value	h_1	$x \bmod 7$	h_2
4371	1	3	4
1323	3	0	7
6173	3	6	1
4199	9	6	1
4344	4	4	3
9679	9	5	2
1989	9		6

Now insert

4371 $\rightarrow$ 1

1323 $\rightarrow$ 3

6173 $\rightarrow$ 3 $\rightarrow$ occupied $\rightarrow 3+1=4$

4199 $\rightarrow$ 9

4344 $\rightarrow$ 4 occupied $\rightarrow 4+3=7$

9679 $\rightarrow$ 9 occupied $\rightarrow 1 \rightarrow 3 \rightarrow 5$

for 1989

$h_1 = 9, h_2 = 6$

$i = 0 \rightarrow 9$

$i = 1 \rightarrow (9+6) \bmod 10 = 5$

$i = 2 \rightarrow (9+12) \bmod 10 = 1$

$i = 3 \rightarrow (9+18) \bmod 10 = 7$

$i = 4 \rightarrow (9+24) \bmod 10 = 3$

$i = 5 \rightarrow (9+30) \bmod 10 = 9$

it can't be insert in is keep cycling through the same position.

Value

Position

Item value

Value

0

-

1

1584

1

4371

3

8581

2

-

3

8581

3

1323

3

8581

4

6173

4

4484

5

9679

4

8581

6

-

5

8581

7

4344

5

8581

8

-

6

8581

9

4199

6

8581

Amir 19/8/20

Example